

Data leak worksheet

Incident summary: A sales manager shared access to a folder of internal-only documents with their team during a meeting. The folder contained files associated with a new product that has not been publicly announced. It also included customer analytics and promotional materials. After the meeting, the manager did not revoke access to the internal folder, but warned the team to wait for approval before sharing the promotional materials with others.

During a video call with a business partner, a member of the sales team forgot the warning from their manager. The sales representative intended to share a link to the promotional materials so that the business partner could circulate the materials to their customers. However, the sales representative accidentally shared a link to the internal folder instead. Later, the business partner posted the link on their company's social media page assuming that it was the promotional materials.

Control	Least privilege
Issue(s)	<i>What factors contributed to the information leak?</i> The sales manager did not follow the principle of least privilege by allowing continuous access to sensitive information, when only the promotional materials needed to be accessed after the meeting. This led to an easily avoidable mistake that could have been prevented by following standard file permission handling.
Review	<i>What does NIST SP 800-53: AC-6 address?</i> This section of the NIST cybersecurity framework addresses the principle of least privilege and how such standards should be implemented within a company's internal network. The most important of these in this instance is only granting read/write/execute privileges to those who absolutely need them, and regularly performing audits to

	ensure that users or outsiders don't have access to privileged information.
Recommendation(s)	<i>How might the principle of least privilege be improved at the company?</i> File permissions should regularly be monitored and changed to reflect the proper level of access for employees, customers, business partners etc. It should not be possible for a simple user error to result in a data leak containing sensitive information regarding company operations. Additionally, files that need to be shared only for a short time should automatically revoke privileges after a set amount of time to ensure that the principle of least privilege is upheld at all times.
Justification	<i>How might these improvements address the issues?</i> These improvements would guard against similar incidents in the future by ensuring that individuals or groups can only access data intended for their use, and only for as long as needed. By continuously examining who has access to what data, the company's security posture is improved and the risk of data leaks is reduced.

Security plan snapshot

The NIST Cybersecurity Framework (CSF) uses a hierarchical, tree-like structure to organize information. From left to right, it describes a broad security function, then becomes more specific as it branches out to a category, subcategory, and individual security controls.

Function	Category	Subcategory	Reference(s)
Protect	PR.DS: <i>Data security</i>	PR.DS-5: <i>Protections against data leaks.</i>	NIST SP 800-53: AC-6

In this example, the implemented controls that are used by the manufacturer to protect against data leaks are defined in NIST SP 800-53—a set of guidelines for securing the privacy of information systems.

Note: References are commonly hyperlinked to the guidelines or regulations they relate to. This makes it easy to learn more about how a particular control should be implemented. It's common to find multiple links to different sources in the references columns.

NIST SP 800-53: AC-6

NIST developed SP 800-53 to provide businesses with a customizable information privacy plan. It's a comprehensive resource that describes a wide range of control categories. Each control provides a few key pieces of information:

- **Control:** A definition of the security control.
- **Discussion:** A description of how the control should be implemented.
- **Control enhancements:** A list of suggestions to improve the effectiveness of the control.

AC-6	Least Privilege
	Control:

	Only the minimal access and authorization required to complete a task or function should be provided to users.
	Discussion: Processes, user accounts, and roles should be enforced as necessary to achieve least privilege. The intention is to prevent a user from operating at privilege levels higher than what is necessary to accomplish business objectives.
	Control enhancements: • Restrict access to sensitive resources based on user role. • Automatically revoke access to information after a period of time. • Keep activity logs of provisioned user accounts. • Regularly audit user privileges.

Note: In the category of access controls, SP 800-53 lists least privilege sixth, i.e. AC-6.